

The Art of Distillation

A Journey into Automated Text Summarization

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Contents

1	Introduction: The Information Overload Era	2
2	Part 1: The Taxonomy of Summarization	2
2.1	1. Output: Extractive vs. Abstractive	2
2.2	2. Input: Single vs. Multi-Document	2
2.3	3. Focus: Generic vs. Query-Focused	3
3	Part 2: The Three-Act Play of Extractive Summarization	3
3.1	Act 1: Content Selection (Who makes the cut?)	3
3.1.1	Approach A: The Frequency Method (Luhn's Idea, 1958)	3
3.1.2	Approach B: Graph-Based (LexRank)	3
3.1.3	The Problem of Redundancy: MMR	4
3.2	Act 2: Information Ordering (Telling a Coherent Story)	4
3.3	Act 3: Sentence Realization (Polishing the Diamond)	5
4	Part 3: The Neural Revolution (Abstractive Summarization)	5
4.1	From RNNs to Sequence-to-Sequence	5
5	Part 4: Evaluating Summaries - ROUGE	6
6	Conclusion	6

1 Introduction: The Information Overload Era

The TL;DR Problem

Imagine you are an executive waking up to 150 unread emails, a 60-page industry report, and the transcripts of three meetings you missed yesterday. You simply do not have the time to read every single word. You desperately need someone to give you the **TL;DR** (Too Long; Didn't Read).

You need the core facts, the main arguments, and the final decisions. Humans do this naturally—we read a 400-page novel and can explain the plot to a friend in three minutes. For a computer, however, reading thousands of words and mathematically deciding which ones are "important" is an incredibly complex NLP challenge.

Text Summarization is the process of producing an abridged (shortened) version of a text while strictly retaining the key, relevant information.

Applications are everywhere today:

- **Search Engines:** Google creating a 26-word snippet to answer "What is Die Brücke?".
- **Corporate Tools:** Generating action items from chat logs or emails.
- **News Aggregators:** Condensing daily news into a 5-minute digest.
- **Accessibility:** Simplifying complex medical or legal texts for laymen.

2 Part 1: The Taxonomy of Summarization

Before we build a summarizer, we need to decide what *kind* of summary we want. Summarization is classified along three major dimensions.

2.1 1. Output: Extractive vs. Abstractive

Analogy Time: The Highlighter vs. The Pen

Extractive Summarization is like giving a student a **yellow highlighter**. They must read a textbook and highlight the most important sentences. The final summary consists *only* of exact sentences lifted directly from the original text. No new words are invented.

Abstractive Summarization is like giving a student a **blank piece of paper and a pen**. They read the textbook, close it, and write down the summary in their own words, potentially using vocabulary that never appeared in the original text.

2.2 2. Input: Single vs. Multi-Document

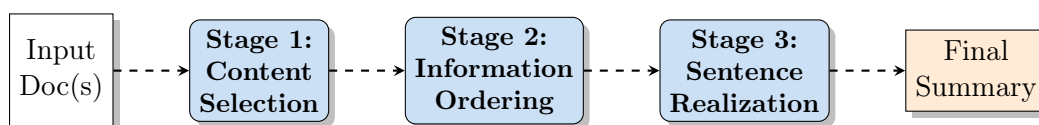
- **Single Document (SDS):** Summarizing one specific text (e.g., generating a headline for a news article or an abstract for a research paper). A surprisingly strong baseline for news SDS is simply extracting the very first sentence!
- **Multi-Document (MDS):** Summarizing a cluster of texts on the same topic (e.g., gathering 5 different news reports about an earthquake and writing one cohesive overview). MDS is much harder because it must handle overlapping facts, contradictory information, and heavy redundancy.

2.3 3. Focus: Generic vs. Query-Focused

- **Generic:** Summarize the whole document broadly. (e.g., "What is this Wikipedia page about?")
- **Query-Focused:** Summarize the document *strictly based on a user's specific question*. (e.g., "Was cast-metal movable type invented in Korea?"). The system acts as a complex question-answering tool, ignoring text that doesn't answer the prompt.

3 Part 2: The Three-Act Play of Extractive Summarization

Extractive summarization is historically the most reliable method because it is immune to "hallucinations" (inventing fake facts). It follows a strict three-stage pipeline.



3.1 Act 1: Content Selection (Who makes the cut?)

In this stage, we segment the document into sentences and give each sentence a "score." The highest-scoring sentences will be selected. But how do we mathematically define "important"?

3.1.1 Approach A: The Frequency Method (Luhn's Idea, 1958)

If a document is about "Mars," the words "Mars," "Rover," and "Space" will appear frequently. We use **TF-IDF** (Term Frequency-Inverse Document Frequency) to find words that are frequent in this document but rare globally.

$$\text{weight}(w_i) = \text{TF}_{ij} \times \text{IDF}_i$$

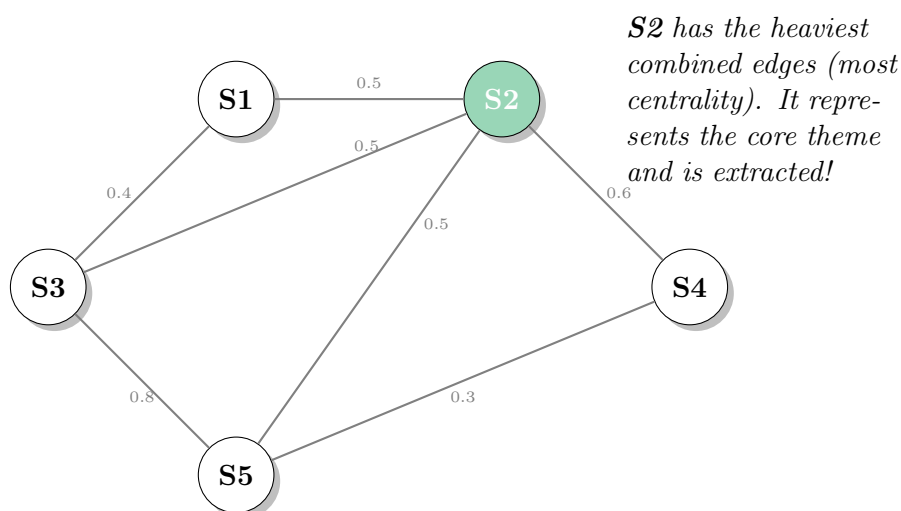
A sentence gets a high score if it contains many high-weight, salient words.

3.1.2 Approach B: Graph-Based (LexRank)

Analogy Time: The Cocktail Party of Sentences

Imagine every sentence in a document is a person at a cocktail party. People (sentences) "talk" to each other if they share similar words. If one person's ideas are agreed with by almost everyone else at the party, that person is the most central figure.

LexRank treats sentences as nodes in a graph. Edges are drawn between sentences that are similar (using Cosine Similarity). We then run an algorithm identical to Google's **PageRank** to find the most "central" sentences!



3.1.3 The Problem of Redundancy: MMR

Imagine we are summarizing multiple articles about J.K. Rowling winning a lawsuit.

- *Sentence 1:* Author J.K. Rowling won her legal battle to ban an unofficial encyclopedia. (Score: 0.9)
- *Sentence 2:* A US judge ruled in favor of J.K. Rowling against the encyclopedia's publication. (Score: 0.88)

If we just take the top 2 highest-scoring sentences, our summary is highly redundant. We need variety!

Key Concept: Maximal Marginal Relevance (MMR)

MMR solves redundancy by balancing two forces: **Relevance** (match to the query/topic) and **Novelty** (difference from sentences already chosen).

$$\hat{S}_{MMR} = \arg \max_{s \in D} \left[\lambda \cdot \text{Sim}(s, Q) - (1 - \lambda) \cdot \max_{s_j \in S} \text{Sim}(s, s_j) \right]$$

- $\lambda \cdot \text{Sim}(s, Q)$: Be as relevant as possible to the query Q .
- $-(1 - \lambda) \cdot \max_{s_j \in S} \text{Sim}(s, s_j)$: Penalty! Don't be too similar to what is already in the summary S !

We iteratively add the best scoring sentence to the summary until we reach our desired length limit K . This is a greedy approach to **Global Optimization**.

3.2 Act 2: Information Ordering (Telling a Coherent Story)

Once we have our extracted sentences, in what order do we present them? If they come from 5 different documents, a random ordering will sound like a madman talking.

- **Chronological:** Order by the date of the source document (great for news).
- **Coherence:** Group sentences so that neighboring sentences share similar entities or have high cosine similarity.
- **Topical Ordering (Information Extraction):** For specific types of queries, we enforce a template:

Question Type	Information Flow
Biography	Dates → Nationality → Education → Fame
Definition	Genus → Species → Synonym → Subtype
Medical/Drug	Population → Problem → Intervention → Outcome

3.3 Act 3: Sentence Realization (Polishing the Diamond)

Extracted sentences are often too long or contain broken references. We must clean them.

1. **Simplification (Pruning):** Strip out unnecessary clauses (appositives, attribution clauses, initial adverbials).
 - *Original:* "Rebels agreed to talks with government officials, international observers said Tuesday."
 - *Pruned:* "Rebels agreed to talks with government officials." (Removed attribution clause).
2. **Context Fixes (Anaphora Resolution):** If our extracted sentence starts with "He", the reader won't know who "He" is.
 - *Original:* "He has immunity from prosecution..."
 - *Fixed:* "Gen. Augusto Pinochet has immunity from prosecution..."

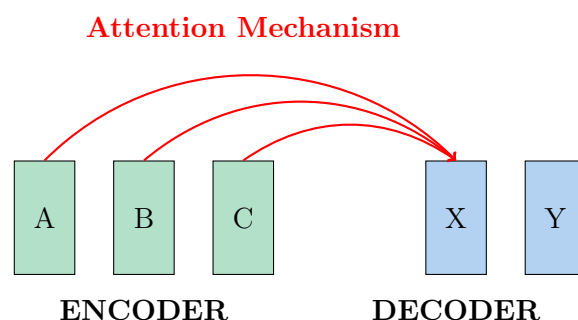
4 Part 3: The Neural Revolution (Abstractive Summarization)

Modern AI has shifted heavily toward **Abstractive** Summarization using Deep Learning, generating completely novel sentences.

4.1 From RNNs to Sequence-to-Sequence

Initially, Recurrent Neural Networks (RNNs) were used, but they had trouble remembering long-range dependencies (they would forget the beginning of the document by the time they reached the end).

The breakthrough was **Sequence-to-Sequence (Seq2Seq) with Attention**.



The **Encoder** reads the entire input document and creates an internal representation. The **Decoder** then writes the summary one word at a time. The **Attention Mechanism** acts as a spotlight, allowing the decoder to "look back" at specific, highly relevant words in the input document while generating the next word in the summary.

Today, this is powered by the **Transformer** architecture, which relies entirely on Multi-Head Attention blocks, enabling state-of-the-art abstractive models. During training, the data is structured as: **ARTICLE TEXT <EOS> SUMMARY <EOS> <pad>**. The model calculates loss only when predicting the summary portion.

5 Part 4: Evaluating Summaries - ROUGE

How do we mathematically prove an AI-generated summary is "good"? We use **ROUGE** (Recall-Oriented Understudy for Gisting Evaluation).

Key Concept: The ROUGE Metric

ROUGE measures how much the machine-generated summary overlaps with reference summaries written by human experts. It primarily looks at **n-gram overlap** (e.g., bigrams - pairs of words).

$$\text{ROUGE-N} = \frac{\sum_{S \in \{\text{RefSummaries}\}} \sum_{\text{n-grams } i \in S} \min(\text{count}(i, X), \text{count}(i, S))}{\sum_{S \in \{\text{RefSummaries}\}} \sum_{\text{n-grams } i \in S} \text{count}(i, S)}$$

Intuition: "Out of all the n-word phrases the human experts used, what percentage did the machine successfully manage to include in its output (X)?"

Example: Calculating ROUGE-2 (Bigrams)

Let's evaluate a system answering: "What is water spinach?"

- **Human 1:** *Water spinach is* a green leafy vegetable grown in the tropics. (10 bigrams)
- **Human 2:** *Water spinach is* a semi-aquatic tropical plant grown as a vegetable. (9 bigrams)
- **Human 3:** *Water spinach is* a commonly eaten leaf vegetable of Asia. (9 bigrams)

Total reference bigrams = 10 + 9 + 9 = **28**.

System output (X): *Water spinach is* a leaf vegetable commonly eaten in tropical areas of Asia.

- Matches with Human 1: 3
- Matches with Human 2: 3
- Matches with Human 3: 6

$$\text{ROUGE-2 Score} = \frac{3 + 3 + 6}{28} = \frac{12}{28} = \mathbf{0.43 \text{ (or 43\%)}}$$

While ROUGE isn't perfect (it struggles to reward valid synonyms in abstractive summaries), it remains the industry standard for rapidly evaluating summarization models.

6 Conclusion

Text Summarization is a beautiful intersection of linguistic rules and mathematical optimization.

- We define summaries by their output (Extractive vs. Abstractive), input size, and query focus.
- Extractive methods mathematically highlight text based on frequency (TF-IDF), centrality (LexRank), and diversity (MMR).
- Deep Learning (Transformers and Attention) allows machines to abtractively paraphrase text, unlocking human-like fluency.
- Automated metrics like ROUGE allow us to evaluate and improve these models continuously.